

Dhruv Thippasandra

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  Dhruv-Thippasandra

Education

University of Southern California

Aug 2025 – May 2027 (On track)

Master of Science in [Electrical Engineering](#)(Data Science and Machine Learning) 

- Relevant Coursework: Probability for Electrical and Computer Engineers, Computational Deep Learning, Machine Learning: Supervised Methods, Database Systems

University of California, Riverside

Sep 2022 – Jun 2025

Bachelor of Science in [Electrical Engineering](#) 

- GPA: 3.77/4.0
- Relevant Coursework: Machine Learning & Data Mining, Digital Signal Processing, Probability, Random Variables & Random Processes in ECE, Electronic Circuits I & II, Signals and Systems I & II, Embedded Systems, Automatic Control.
- Honors: Graduated **Cum Laude**; **Chancellor's Honors List** (all years); **Dean's Honors List** (all quarters)

Work Experience

Software Engineering Intern

Jun 2026 - Aug 2026

AT&T

- Engineered a scalable backend anomaly detection service using **asynchronous FastAPI endpoints**, integrating an **Extended Isolation Forest (EIF)** model to automatically detect and suspend telecom fraud.
- Built automated **Azure ETL pipelines** to ingest **12M+ phone records** from Snowflake to PostgreSQL, executing bi-hourly incremental upserts of user telemetry metrics to trigger batch fraud inference.
- Designed a **real-time decision engine** that auto-suspends fraudulent users hitting an optimized 0.7 anomaly threshold; engineered a monthly MLOps pipeline to retrain models on the latest **5M+ timestamped records**, promoting the newly trained model to production only upon PU F-score improvement.

Machine Learning Engineer

Sep 2025 - Jan 2026

Mercor

- Compiled external ML competitions into challenging tasks reflecting real-world ML responsibilities. Drafted detailed natural-language execution plans and implemented them in Python within a Docker environment, validating implementations against original plans and marking discrepancies. These tasks are used by researchers to train models on high-quality data.

Projects

Network Degradation Prediction Using Real Internet Telemetry

2025 – 2026 School year

- Trained a SMOTE-balanced XGBoost classifier on **327MB** of real RIPE Atlas telemetry (Root DNS anchor) to predict network degradation **5 minutes** ahead, achieving **91% accuracy** and **40% recall** on a severely imbalanced <1% minority class via adaptive Z-score labeling.
- Deployed on **AWS EC2** with Docker and **CI/CD** via GitHub Actions; decoupled inference from serving via cron-based background refresh, achieving **sub-100ms** page loads on a t3.micro with zero model cold-start latency.

Apollo: Autonomous AI Research Analyst

2025 – 2026 School year

- Built an autonomous Python pipeline to ingest arXiv cs.AI papers biweekly, applying a **three-layer filtering system** (keyword scoring, author boosting, Gemma 3 27B LLM evaluation) to surface the **top 25 most impactful** from ~1500 submissions per cycle, auto-deployed as GitHub Pages digests.
- Integrated a **hybrid SQLite vector+keyword store** for persistent RAG memory and deployed a Telegram bot backed by **Gemini 2.5 Flash** for real-time grounded querying of the curated papers.

Wildfire Spread Forecaster Hosted on Proxmox Home-Cloud

2025 – 2026 School year

- Architected a **3-node bare-metal Proxmox cluster** with VFIO GPU passthrough (RTX 3080, GTX 1060, RTX 3050), orchestrated via Nomad (lightweight K8s alternative); replaced **~\$400/mo** in equivalent cloud GPU instances, exposing 2 production services([dhruvq.com](#) , [fire.dhruvq.com](#) ) via Cloudflare Tunnels + Tailscale with zero open inbound ports.
- Engineered an async **FastAPI + Next.js** prediction engine using a Rothermel-inspired Cellular Automata algorithm on a **100m grid**, fusing live NASA FIRMS, NOAA HRRR, and USGS 3DEP data to forecast fire spread **up to 8 hours** from start; Validated via backtest simulation, achieving **75.4% IoU** against modeled ground-truth fire perimeters.

Skills

Languages & Frameworks: Python, SQL, TypeScript, JavaScript, FastAPI, Next.js, Node.js, Snowflake, PostgreSQL

ML & Infrastructure: PyTorch, XGBoost, MLOps, Azure, AWS, Docker, Nomad, Proxmox, Tailscale, Cloudflare, GitHub Actions, Linux